

LMS Testing Lifecycle Project: SDLC phases Waterfall vs Agile.

Project: Learning Management System (LMS)

The LMS allows:

- Students to register and login
- Students to search and enroll in courses
- Students to view course content
- Admins to manage courses

Requirements

Let's start with only 2 requirements.

- Requirement 1 (R1)

Student Login

As a student, I want to log in to the LMS so that I can access my courses.

- Requirement 2 (R2)

Course Enrollment

As a student, I want to enroll in a course so that I can start learning.

SDLC

Waterfall Model	Agile Model
Requirement Gathering Deliverables • BRD • Functional Requirements Requirements gathered: • R1: Student Login	Product Backlog IDUser Story US1Student Login US2Course Enrollment

- R2: Course Enrollment

QA Activities

- Review requirements
- Identify ambiguities
- Prepare questions

Design Phase

Deliverables

- UI Mockups
- Database Design

QA Activities

- Review design documents
- Identify testable components

Development Phase

Developers build:

Login Module

- Username field
- Password field
- Login button

Enrollment Module

- Course list
- Enroll button

QA Activities

- Create Test Scenarios
- Create Test Cases

Testing Phase

QA executes tests.

Login Tests

Sprint 1

User Story

US1 - Student Login

Sprint Activities

- Development
- Testing
- Bug Fixing

Deliverables

- Login Feature

Sprint 2

User Story

US2 - Course Enrollment

Sprint Activities

- Development
- Testing
- Bug Fixing

Deliverables

- Enrollment Feature

Daily Scrum

Example:

Yesterday:
Created login test cases

Today:
Execute login tests

Blockers:
None

- Valid Login
- Invalid Login
- Empty Username
- Empty Password

Enrollment Tests

- Successful Enrollment
- Duplicate Enrollment
- Course Full

Deployment Phase

Application released.

QA Activities

- Smoke Testing

Example:

- Login works
- Enrollment works

Maintenance Phase

Developers fix defects.

QA Activities

- Regression Testing
- Retesting

Sprint Review

Demonstrate:

- Login feature completed
- Test cases executed

Sprint Retrospective

What went well:

Requirements were clear

What can improve:

Need earlier QA involvement